SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Salt-and-Pepper Noise
A scanned document contains salt-and-pepper noise affecting readability.
(a) Explain the scenario and characteristics of this noise.
(b) Identify key issues impacting image quality.
(c) Apply an appropriate filtering technique.
(d) Justify why this method is more suitable than alternatives.
(e) Present your response clearly and logically.

2. Object Boundary Detection
In a medical image, boundaries between tissues need to be clearly identified.
(a) Interpret the scenario and explain the importance of boundary detection.
(b) Identify key challenges in detecting boundaries.
(c) Apply a suitable edge detection method.
(d) Justify your selection with logical reasoning.
(e) Present your explanation clearly and systematically.

3. Simple Segmentation
An image contains objects clearly distinguishable from the background based on intensity.
(a) Interpret the scenario and explain segmentation requirements.
(b) Identify key features enabling separation.
(c) Apply a suitable segmentation technique.
(d) Justify your decision logically.
(e) Present your response clearly.

4. Low Brightness Image

An image captured in low light appears very dark with poor visibility.
(a) Interpret the scenario and explain the cause of low brightness.
(b) Identify key issues affecting image visibility.
(c) Apply an appropriate enhancement technique.
(d) Justify your choice with reasoning.
(e) Present your explanation clearly.

5. Gaussian Noise

An image contains Gaussian noise affecting smooth regions.
(a) Interpret the scenario and describe Gaussian noise.
(b) Identify key issues affecting quality.
(c) Apply an appropriate smoothing filter.
(d) Justify your method over alternatives.
(e) Present your explanation clearly.

6. Loss of Fine Details

An image loses fine details after processing.
(a) Interpret the scenario and explain detail loss.
(b) Identify key issues related to filtering.
(c) Apply a suitable enhancement technique.
(d) Justify your method with reasoning.
(e) Present your explanation clearly.

7. Edge Enhancement Requirement

An application requires highlighting edges for feature extraction.
(a) Interpret the scenario and explain the need for edge enhancement.
(b) Identify key issues in detecting edges.
(c) Apply an appropriate method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CO2 AT1 SCENARIO BASED QUESTIONS
ITA0509-COMPUTER VISION

1. Salt-and-Pepper Noise

(a) Scenario and Characteristics

A scanned document contains many small black and white dots because of dust, faulty scanners, or transmission errors. This type of noise is called Salt-and-Pepper Noise. White pixels are called "salt" and black pixels are called "pepper." It reduces the readability of the document.

(b) Key Issues

- Black and white dots hide the text.
- Image becomes difficult to read.
- Characters may appear broken.
- OCR (Optical Character Recognition) accuracy decreases.

(c) Appropriate Filtering Technique

Median Filter

Working:

The median filter replaces each noisy pixel with the median value of its neighboring pixels. This removes isolated noisy pixels while preserving edges.

(d) Justification

The Median Filter is the best choice because:

- Removes salt-and-pepper noise effectively.
- Preserves edges and text.
- Does not blur the image like the Mean Filter.
- Produces a cleaner document for reading and OCR.

(e) Conclusion

Therefore, the Median Filter is the most suitable method for removing salt-and-pepper noise and improving document quality.

2. Object Boundary Detection

(a) Scenario

In a medical image, doctors need to identify the boundaries between different tissues or organs. Accurate boundary detection helps in disease diagnosis, tumor detection, and surgical planning.

(b) Key Challenges

- Weak edges.
- Image noise.
- Low contrast.
- Similar intensity between tissues.

(c) Suitable Edge Detection Method

Canny Edge Detection

Working

1. Removes noise using Gaussian Filter.
2. Calculates image gradients.
3. Performs Non-Maximum Suppression.
4. Uses Double Thresholding.
5. Tracks strong edges.

(d) Justification

Canny Edge Detection is preferred because:

- Produces accurate edges.
- Detects weak and strong edges.
- Removes false edges caused by noise.

- Gives thin and continuous boundaries.

(e) Conclusion

Therefore, Canny Edge Detection is the best method for medical image boundary detection.

3. Simple Segmentation

(a) Scenario

The objects in the image have different brightness compared to the background, making them easy to separate.

(b) Key Features

- Large intensity difference.
- Clear contrast.
- Simple background.
- Uniform object color.

(c) Suitable Segmentation Technique

Threshold Segmentation

Working

A threshold value is selected.

- Pixels greater than the threshold become object.
- Pixels lower than the threshold become background.

(d) Justification

Thresholding is suitable because:

- Easy to implement.
- Fast processing.
- High accuracy when object and background have different intensities.
- Requires less computation.

(e) Conclusion

Threshold Segmentation is the most efficient method for separating objects from the background.

4. Low Brightness Image

(a) Scenario

The image is captured under poor lighting conditions, making it dark and difficult to see.

(b) Key Issues

- Low visibility.
- Hidden image details.
- Poor contrast.
- Difficult object recognition.

(c) Appropriate Enhancement Technique

Histogram Equalization

Working

Histogram Equalization redistributes pixel intensity values across the image, increasing overall brightness and contrast.

(d) Justification

Histogram Equalization is suitable because:

- Improves brightness.
- Enhances contrast.
- Reveals hidden details.
- Makes the image easier to analyze.

(e) Conclusion

Histogram Equalization significantly improves visibility in low-light images.

5. Gaussian Noise

(a) Scenario

The image contains random intensity variations caused by sensors or electronic devices. This is called Gaussian Noise.

(b) Key Issues

- Grainy appearance.
- Reduced image quality.
- Loss of smooth regions.
- Lower analysis accuracy.

(c) Appropriate Smoothing Filter

Gaussian Filter

Working

The Gaussian Filter smooths the image using a Gaussian kernel, reducing random noise while preserving important structures.

(d) Justification

Gaussian Filter is preferred because:

- Specifically designed for Gaussian noise.
- Smooths images naturally.
- Reduces random fluctuations.
- Better than Median Filter for Gaussian noise.

(e) Conclusion

The Gaussian Filter effectively removes Gaussian noise while maintaining image quality.

6. Loss of Fine Details

(a) Scenario

After filtering, the image appears smooth but small details and textures are missing.

(b) Key Issues

- Edge blurring.
- Texture loss.
- Reduced sharpness.
- Lower image clarity.

(c) Suitable Enhancement Technique

Unsharp Masking (Image Sharpening)

Working

The blurred image is subtracted from the original image, and the difference is added back to sharpen edges.

(d) Justification

Unsharp Masking is suitable because:

- Restores fine details.
- Sharpens edges.
- Improves clarity.
- Better than increasing brightness, which cannot recover lost details.

(e) Conclusion

Unsharp Masking effectively restores sharpness and fine image details.

7. Edge Enhancement Requirement

(a) Scenario

The application requires highlighting object boundaries before feature extraction.

(b) Key Issues

- Weak edges.
- Noise interference.
- Low contrast.

- Missing object boundaries.

(c) Appropriate Method

Sobel Edge Detection

Working

The Sobel operator calculates horizontal and vertical intensity gradients to highlight edges.

(d) Justification

Sobel is preferred because:

- Simple implementation.
- Detects horizontal and vertical edges.
- Faster than Canny.
- Suitable for feature extraction.

(e) Conclusion

Sobel Edge Detection effectively enhances edges for feature extraction.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand